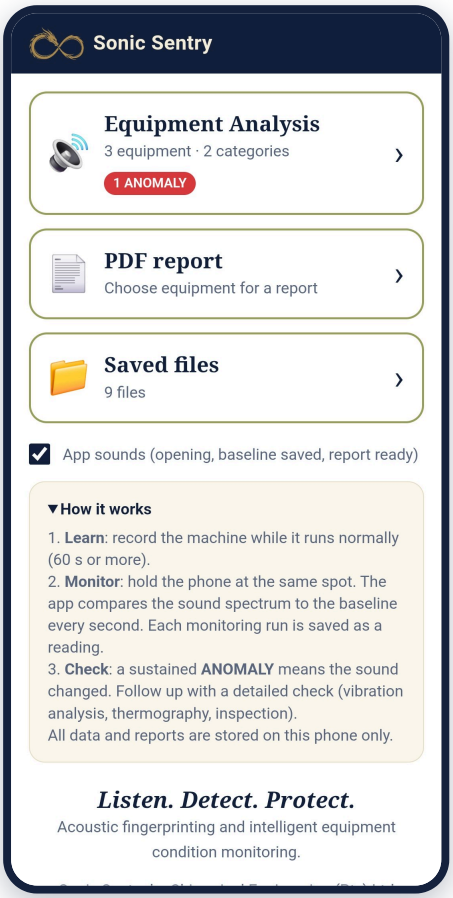
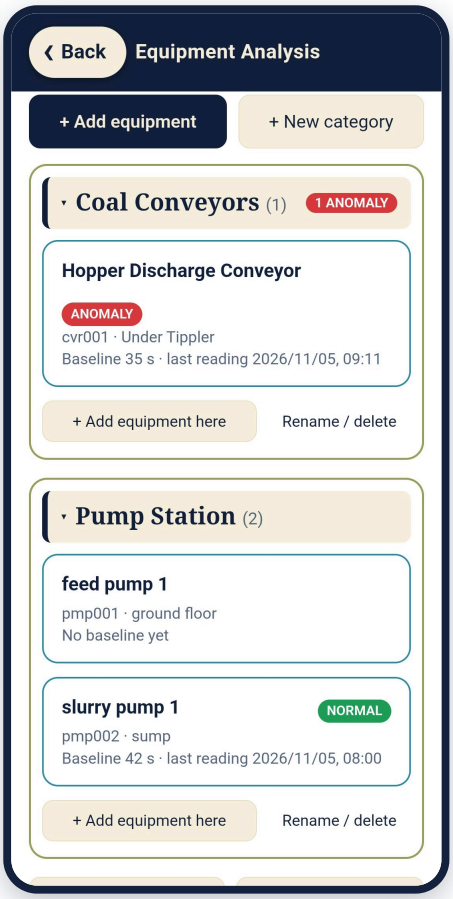
Machines often sound different before they fail.

Sonic Sentry turns an Android phone into an acoustic screening tool. It learns how each machine sounds when healthy, then shows you when that sound changes.

Walk-down rounds depend on experienced ears, and that experience is hard to pass between shifts. Sonic Sentry gives every technician the same reference: a saved sound signature for each machine.







Starts with a phone

No sensors to fit and no cabling. Use the phone microphone, or plug in an external microphone.

Shows what changed

See which frequency range moved, by how much, and what that range usually points to.

Keeps data with you

No accounts and no internet needed. No audio is recorded, and nothing leaves the phone unless you share it.

Three steps from sound to decision

The app measures 60 frequency bands from 20 Hz to 20 kHz and compares them with the machine's baseline every second.

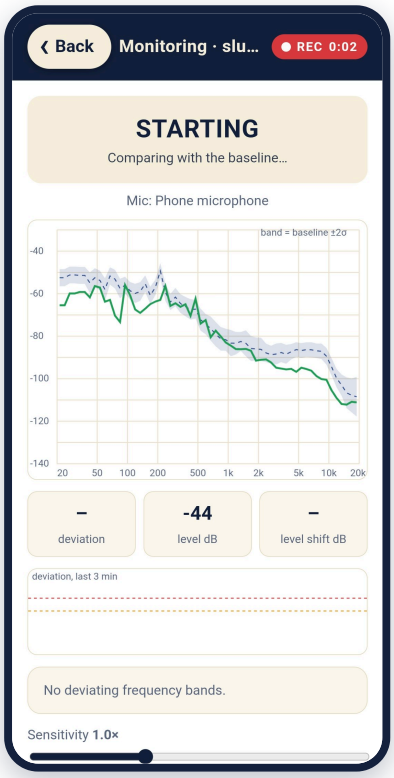
1 Learn

Hold the phone at a fixed spot for 60 seconds or more while the machine runs normally. Its sound spectrum becomes the baseline.



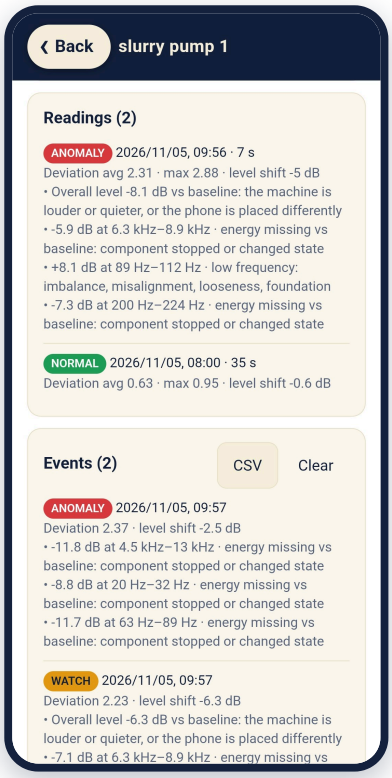
2 Monitor

Return to the same spot on your next round. The live spectrum is drawn over the normal band, so a change is easy to see.



3 Check

Every run is saved as a reading. Findings list the changed ranges, so you know where to aim a detailed check.



Clear status

- **NORMAL** Sound matches the baseline.
- **WATCH** A sustained change. Keep an eye on it.
- **ANOMALY** A larger sustained change. Plan a check.

A change must last 3 seconds, so a single bang does not raise a flag.

What a louder range usually points to

Below 50 Hz	Slow-speed imbalance, structural or foundation movement, pulsation
50–200 Hz	Imbalance, misalignment, looseness, foundation
200 Hz–1 kHz	Gear mesh, belts, flow or pump noise, structural resonance
1–4 kHz	Bearing wear, gear damage, cavitation, rubbing
Above 4 kHz	Bearing or lubrication problems, air or steam leaks, electrical arcing

Where it earns its place

Use it on machines that run steadily and have no fixed sensors. A change in sound flags where to look next.

Materials handling

- **Belt conveyors:** failing idlers, belt rubbing, worn pulley bearings
- **Drive units:** gearboxes and motors on conveyors and bucket elevators
- **Screw and chain conveyors:** rubbing, worn links, load changes
- **Crushers and screens:** drive bearing wear and looseness
- **Stackers, reclaimers, tipplers:** drives and hydraulic power packs

Fluid handling

- **Pumps:** cavitation, bearing wear and flow changes on slurry, process and water duties
- **Compressors:** valve and bearing changes, air leaks
- **Fans and blowers:** imbalance from build-up, bearing wear
- **Valves and steam traps:** passing valves, steam or air leaks
- **Agitators and mixers:** gearbox and bearing changes

Rotating process equipment

- **Rotary kilns and dryers:** girth gear and pinion mesh, main drive, support and thrust rollers
- **Ball and SAG mills:** gearbox, pinion and trunnion areas
- **Pulp and paper:** refiners, machine drives, vacuum pumps
- **Centrifuges and decanters:** imbalance and bearing changes

Power and electrical

- **Motors and generators:** bearing and cooling fan noise
- **Transformers:** changes in the normal hum, cooling fans, oil pumps
- **Cooling towers:** fan gearboxes and imbalance
- **Turbine auxiliaries:** lube oil pumps and support equipment

Workshops and mobile plant

- **Machine tools:** spindles and gearboxes on large lathes and presses
- **Rolling mills:** main drives and gearboxes
- **Overhead cranes:** hoist gearboxes and travel wheels
- **Mobile plant:** engines and hydraulic pumps at a fixed engine speed

Across the machine's life

- **Commissioning:** keep a healthy signature from day one
- **After repairs:** record a new baseline after an overhaul
- **Duty and standby units:** track each one against its own baseline
- **Contractor handover:** a PDF record of condition before and after work

Best fit

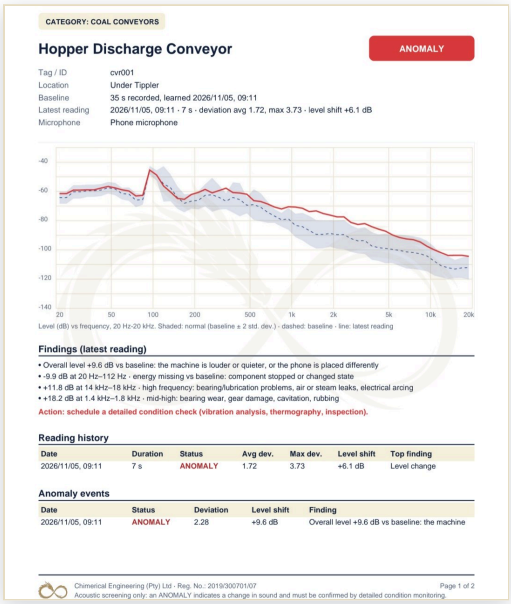
Machines that run steadily in a repeatable operating state. Make a separate entry for each operating mode, such as full load and idle.

Take more care

In loud areas a neighbouring machine can mask a change, so hold the phone close, in the same spot. Very slow parts, like a kiln shell, suit a phone microphone less than their faster drives do.

Reports your team can act on

Tick the equipment you want and create a branded PDF on the phone, with one page per machine.


CATEGORY: COAL CONVEYORS

Hopper Discharge Conveyor — ANOMALY

Tag / ID cvr001
Location Under Tippler
Baseline 35 s recorded, learned 2026/11/05, 09:11
Latest reading 2026/11/05, 09:11 - 7 s - deviation avg 1.72, max 3.73 - level shift +6.1 dB
Microphone Phone microphone

Findings (latest reading)
- Overall level +9.6 dB vs baseline: the machine is louder or quieter, or the phone is placed differently
- -9.9 dB at 20 Hz–112 Hz - energy missing vs baseline: component stopped or changed state
- +11.8 dB at 14 kHz–18 kHz - high frequency: bearing/lubrication problems, air or steam leaks, electrical arcing
- +18.2 dB at 1.4 kHz–1.8 kHz - mid-high: bearing wear, gear damage, cavitation, rubbing

Action: schedule a detailed condition check (vibration analysis, thermography, inspection).

Reading history

Date	Duration	Status	Avg dev.	Max dev.	Level shift	Top finding
2026/11/05, 09:11	7 s	ANOMALY	1.72	3.73	+6.1 dB	Level change

Anomaly events

Date	Status	Deviation	Level shift	Finding
2026/11/05, 09:11	ANOMALY	2.28	+9.6 dB	Overall level +9.6 dB vs baseline: the machine

Chimerical Engineering (Pty) Ltd - Reg. No.: 2019/300701/07 — Page 1 of 2
Acoustic screening only: an ANOMALY indicates a change in sound and must be confirmed by detailed condition monitoring.

An anomaly page: spectrum against baseline, findings and a recommended action.


SONIC SENTRY
INDUSTRIAL ACOUSTIC INTELLIGENCE
Listen. Detect. Protect.

ACOUSTIC CONDITION REPORT
Chimerical Engineering (Pty) Ltd
Reg. No.: 2019/300701/07

Acoustic fingerprinting and intelligent equipment condition monitoring.

CATEGORY: PUMP STATION

slurry pump 1 — NORMAL

Tag / ID pmp002
Location sump
Baseline 42 s recorded, learned 2026/11/05, 08:00
Latest reading 2026/11/05, 08:00 - 35 s - deviation avg 0.63, max 0.95 - level shift -0.6 dB
Microphone Phone microphone
Notes inside drain sump

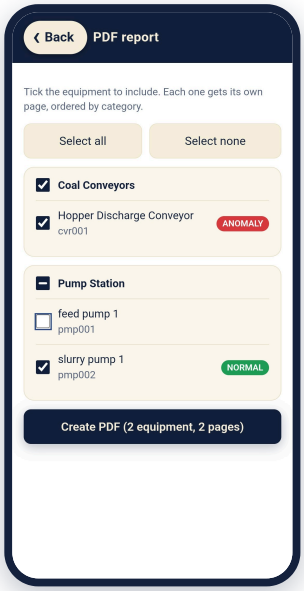
Findings (latest reading)
Sound matched the baseline. No action needed.

Reading history

Date	Duration	Status	Avg dev.	Max dev.	Level shift	Top finding
2026/11/05, 08:00	35 s	NORMAL	0.63	0.95	-0.6 dB	–

Anomaly events
None recorded.

A normal page: a record that the machine matched its baseline.



Built for the plant floor

- Group equipment into your own categories, such as areas or systems
- Status at a glance for every machine and category
- Reading history and an anomaly event log for each machine
- Adjustable sensitivity for quiet or noisy areas
- Silent during monitoring: no sound or vibration to affect the reading
- Saved files sorted by date and filtered by category
- Share by email, WhatsApp or Drive; export events as CSV

Who it is for

Maintenance teams, reliability engineers and plant technicians looking after pumps, conveyors, fans, motors and gearboxes.

A screening tool, not a diagnosis. An ANOMALY means the sound changed and a detailed check is needed (vibration analysis, thermography, inspection). Use the same phone and the same spot each time. Phone microphones are weaker at very low frequencies.

CHIMERICAL ENGINEERING (PTY) LTD
REG. NO.: 2019/300701/07

See it on your own equipment

Ask us for a demonstration of Sonic Sentry.

chimericaleng@gmail.com